

TCX2101: Calculus and Linear Algebra for Computing

Chapter 2: Limits and Continuity

Introduction to Limits

Zeno's paradox

A tortoise and a hare have decided to settle once and for all who is the fastest, so they organize a race to be run over 100 meters. The hare, feeling confident, lets the tortoise have a 50-meter head start.

The starting pistol goes off, and they both start racing toward the finish.

The hare, traveling at twice the speed of the tortoise, reaches the 50-meter mark where the tortoise started.

The tortoise, meanwhile, has traveled only 25 meters, so is at the 75-meter mark.

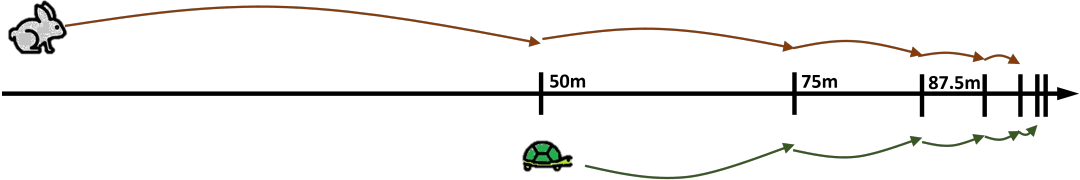
The hare then hops to the 75-meter mark, but the tortoise is still slightly ahead, at the 87.5-meter mark.

The rabbit again hops to where the tortoise was, but in that time, the tortoise has moved a little bit further still.

And so on.

No matter how many times the hare tries to catch up, the tortoise remains slightly ahead. And so, Zeno said, the hare can never overtake the tortoise!

Zeno's paradox



Frog and a Wall

A frog is 1 meters away from a wall.

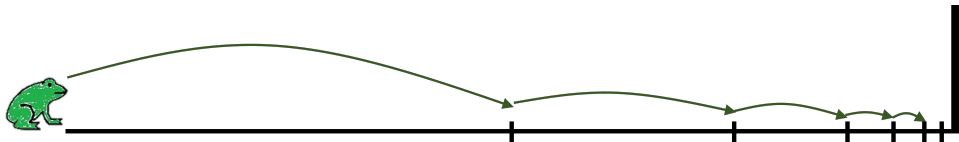
At time $t = 1$, the frog leaps $\frac{1}{2}$ meter towards the wall.

At time $t = 2$, the frog leaps a further $\frac{1}{4}$ meters towards the wall.

At time $t = 3$, the frog leaps a further $\frac{1}{8}$ meters towards the wall.

And so on, that is, at time $t = n$, the frog leaps a further $\frac{1}{2^n}$ meters towards the wall.

Will the frog reach the wall? If it never reach the wall, how near from the wall can the frog be?



The wall is known as the **limit**, that is, the frog can be as near to the wall as possible, but it may not be able to reach it.

Limits by Observing the Graph of a Function

Let

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

For x near 1, but $x \neq 1$, we have

$$f(x) = \frac{(x-1)(x+1)}{x-1} = x+1.$$

- ▶ $f(x)$ can be as near to 2 as we want.
- ▶ $f(x)$ is getting nearer to 2 when x is getting nearer to 1.

We say that the **limit** of $f(x)$ **as x approaches 1** is **2**, denoted as

$$\lim_{x \rightarrow 1} f(x) = 2.$$

Go to [https:](https://www.geogebra.org/calculator)

[//www.geogebra.org/calculator](https://www.geogebra.org/calculator).

- ▶ Enter $f(x) = (x^2 - 1) / (x - 1)$
- ▶ Enter $g(x) = x + 1$

Limits by Observing the Graph of a Function

Let

$$f(x) = \begin{cases} x+1, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

- (i) $\lim_{x \rightarrow 0} f(x) = 1$.
- (ii) $f(0) = 0$.
- (iii) $\lim_{x \rightarrow a} f(x) = a + 1$ for any $a \in \mathbb{R}$.

Go to <https://www.geogebra.org/calculator>.

- ▶ Enter $f(x) = \text{If}(x > 0, x+1, \text{If}(x < 0, x+1, 0))$
- ▶ Enter a
- ▶ Enter $(a, f(a))$

Limits by Observing the Graph of a Function

Let

$$f(x) = \lfloor x \rfloor.$$

- (i) $\lim_{x \rightarrow \frac{1}{2}} f(x) = 0.$
- (ii) $\lim_{x \rightarrow 1.99} f(x) = 1.$
- (iii) $\lim_{x \rightarrow -0.99} f(x) = -1.$

Go to <https://www.geogebra.org/calculator>.

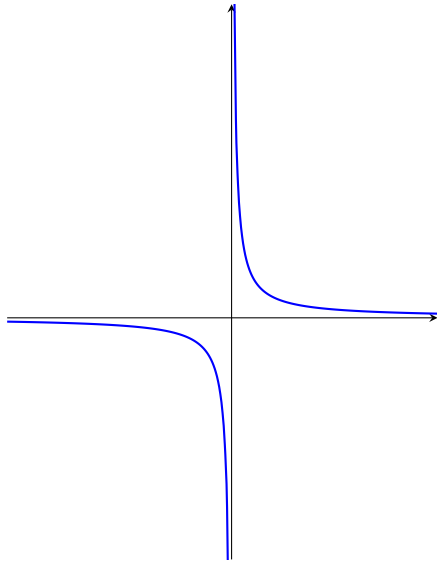
- ▶ Enter $f(x)=\text{floor}(x)$
- ▶ Enter a
- ▶ Enter $(a, f(a))$

Question. What is $\lim_{x \rightarrow 1} f(x)$?

Limits by Observing the Graph of a Function

Is

$$\frac{1}{0} = \infty?$$



Limits by Evaluating a Function

Constant: If $f(x) = c$, then $\lim_{x \rightarrow a} f(x) = c$ for any $a \in \mathbb{R}$.

Identity function: If $f(x) = x$, then $\lim_{x \rightarrow a} f(x) = a$ for any $a \in \mathbb{R}$.

Polynomial: Claim that if $p(x)$ is a polynomial, then $\lim_{x \rightarrow a} p(x) = p(a)$ for any $a \in \mathbb{R}$.

(i) $p(x) = x^2 + 1$

(ii) $p(x) = x^3 - 2x + 2$

Properties of Limits

Limit Laws

Explain, either by the graphs, or by evaluating, the following limit laws.

Suppose

$$\lim_{x \rightarrow c} f(x) = L, \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M.$$

(i) Sum/Difference rule:

$$\lim_{x \rightarrow c} (f(x) \pm g(x)) = L \pm M$$

(ii) Scalar multiple rule:

$$\lim_{x \rightarrow c} (k \cdot f(x)) = k \cdot L$$

(iii) Product rule:

$$\lim_{x \rightarrow c} (f(x) \cdot g(x)) = L \cdot M$$

(iv) Quotient rule:

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$$

Limit Laws

(v) Power rule:

$$\lim_{x \rightarrow c} [f(x)]^n = L^n, \quad n \in \mathbb{Z}^+$$

(vi) Root rule:

$$\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{\frac{1}{n}}, \quad n \in \mathbb{Z}^+,$$

and if n is even, we assume $L \geq 0$.

Corollary

For any *algebraic function* $f(x)$ and any $c \in D_f$,

$$\lim_{x \rightarrow c} f(x) = f(c).$$

Remark

More generally, if $f(x)$ is a **continuous at c** , then $\lim_{x \rightarrow c} f(x) = f(c)$.

Discussion

Prove that $\frac{0}{0} = 0$.

Consider the function $\frac{x^2}{x}$. For $x \neq 0$, $\frac{x^2}{x} = \frac{x}{1} = x$. Hence,

$$\lim_{x \rightarrow 0} \frac{x^2}{x} = \lim_{x \rightarrow 0} x = 0.$$

On the other hand, applying the quotient rule,

$$\lim_{x \rightarrow 0} \frac{x^2}{x} = \frac{\lim_{x \rightarrow 0} x^2}{\lim_{x \rightarrow 0} x} = \frac{0}{0}.$$

Hence,

$$\frac{0}{0} = \lim_{x \rightarrow 0} \frac{x^2}{x} = 0.$$

Question. Is the argument correct? If not, where is the error?

Example

Evaluate the following limits.

1. $\lim_{x \rightarrow -1} \frac{x^3 + 4x^2 - 3}{x^2 + 5}$

$$\lim_{x \rightarrow -1} x^3 + 4x^2 - 3 = (-1)^3 + 4(-1)^2 - 3 = -1 + 4 - 3 = 0, \text{ and}$$

$$\lim_{x \rightarrow -1} x^2 + 5 = (-1)^2 + 5 = 1 + 5 = 6. \text{ So, by quotient rule,}$$

$$\lim_{x \rightarrow -1} \frac{x^3 + 4x^2 - 3}{x^2 + 5} = \frac{\lim_{x \rightarrow -1} x^3 + 4x^2 - 3}{\lim_{x \rightarrow -1} x^2 + 5} = \frac{0}{6} = 0.$$

2. $\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x}$

First note that $\lim_{x \rightarrow 1} x^2 - x = (1)^2 - 1 = 0$. So, we cannot apply quotient rule. However, $x^2 - x = x(x - 1)$ and $x^2 + x - 2 = (x - 1)(x + 2)$. So,

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x} = \lim_{x \rightarrow 1} \frac{(x - 1)(x + 2)}{x(x - 1)} = \lim_{x \rightarrow 1} \frac{x + 2}{x} = \frac{\lim_{x \rightarrow 1} x + 2}{\lim_{x \rightarrow 1} x} = \frac{1 + 2}{1} = 3,$$

where in the second equality, we used $x \neq 1$.

Example

3 $\lim_{x \rightarrow -2} \sqrt{4x^2 - 3}$

$$\lim_{x \rightarrow -2} \sqrt{4x^2 - 3} = \sqrt{\lim_{x \rightarrow -2} (4x^2 - 3)} = \sqrt{4(\lim_{x \rightarrow -2} x^2) - 3} = \sqrt{4(-2)^2 - 3} = \sqrt{16 - 3} = \sqrt{13}.$$

4 $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2}$

$\lim_{x \rightarrow 0} x^2 = 0$, so we cannot use quotient rule. Rationalizing the numerator,

$$\begin{aligned} \frac{\sqrt{x^2 + 100} - 10}{x^2} & \left(\frac{\sqrt{x^2 + 100} + 10}{\sqrt{x^2 + 100} + 10} \right) = \frac{x^2 + 100 - 100}{x^2(\sqrt{x^2 + 100} + 10)} = \frac{1}{\sqrt{x^2 + 100} + 10} \\ \Rightarrow \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2} & = \lim_{x \rightarrow 0} \frac{1}{\sqrt{x^2 + 100} + 10} = \frac{1}{20}. \end{aligned}$$

Limits of Polynomial and Rational Functions

Theorem

If $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$, then

$$\lim_{x \rightarrow c} p(x) = p(c) = a_n c^n + a_{n-1} c^{n-1} + \cdots + a_0.$$

Theorem

If $p(x)$ and $q(x)$ are polynomials with $q(c) \neq 0$, then

$$\lim_{x \rightarrow c} \frac{p(x)}{q(x)} = \frac{p(c)}{q(c)}.$$

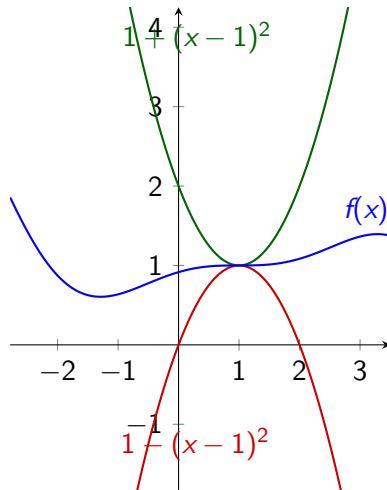
The Sandwich Theorem

Example

Suppose for all x in $(0, 2)$,

$$1 - (x - 1)^2 \leq f(x) \leq 1 + (x - 1)^2.$$

Question. What is $\lim_{x \rightarrow 1} f(x)$?



The Sandwich Theorem

Theorem (The Sandwich Theorem/Squeeze Theorem)

Suppose $g(x) \leq f(x) \leq h(x)$ for all x in some open interval containing c , *except possibly at $x = c$ itself*.

Suppose also that

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L.$$

Then

$$\lim_{x \rightarrow c} f(x) = L.$$

Example

Given a function $u(x)$ that satisfies

$$1 - \frac{x^2}{4} \leq u(x) \leq 1 + \frac{x^2}{2} \quad \text{for all } x \neq 0,$$

find $\lim_{x \rightarrow 0} u(x)$.

Since

$$\lim_{x \rightarrow 0} \left(1 - \frac{x^2}{4}\right) = 1 \quad \text{and} \quad \lim_{x \rightarrow 0} \left(1 + \frac{x^2}{2}\right) = 1,$$

by the sandwich theorem,

$$\lim_{x \rightarrow 0} u(x) = 1.$$

Discussion

What is

$$\lim_{x \rightarrow 0} x^2 \sin \left(\frac{1}{x} \right)?$$

By the product rule,

$$\begin{aligned} \lim_{x \rightarrow 0} x^2 \sin \left(\frac{1}{x} \right) &= \left(\lim_{x \rightarrow 0} x^2 \right) \left(\lim_{x \rightarrow 0} \sin \left(\frac{1}{x} \right) \right) \\ &= 0 \cdot L \\ &= 0. \end{aligned}$$

Question. Is this correct?

Example

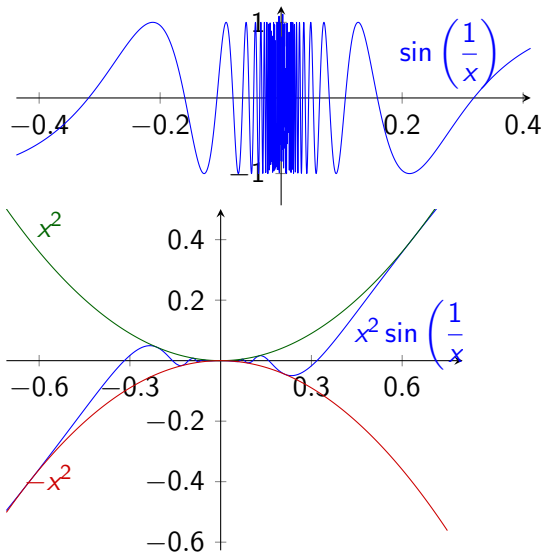
$\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist!

For $x \neq 0$, $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$. Hence,

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2.$$

Also, $\lim_{x \rightarrow 0} -x^2 = \lim_{x \rightarrow 0} x^2 = 0$. Therefore, by **sandwich theorem**,

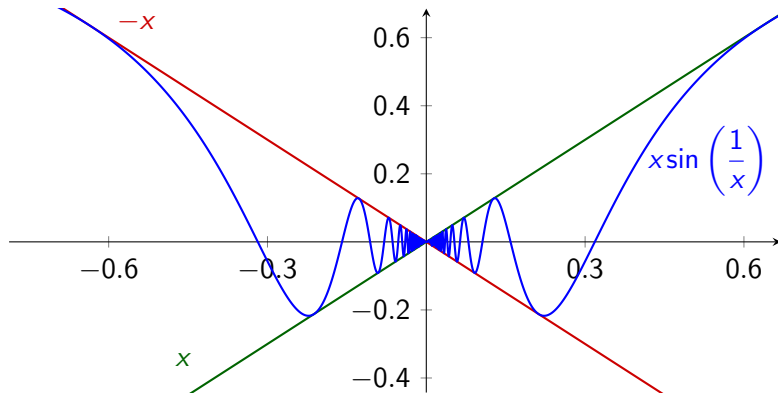
$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0.$$



Challenge

Can we use the sandwich theorem to prove

$$\lim_{x \rightarrow 0} x \sin \left(\frac{1}{x} \right) = 0?$$



$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$$

Theorem

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1.$$

$$\lim_{x \rightarrow 0} \frac{\cos(x) - 1}{x}$$

Corollary

$$\lim_{x \rightarrow 0} \frac{\cos(x) - 1}{x} = 0.$$

Proof.

Using the half angle formula, $\cos(x) = 1 - 2 \sin^2\left(\frac{x}{2}\right)$,

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\cos(x) - 1}{x} &= \lim_{x \rightarrow 0} -\frac{2 \sin^2\left(\frac{x}{2}\right)}{x} = -\lim_{x \rightarrow 0} \left(\frac{\sin\left(\frac{x}{2}\right)}{\frac{x}{2}} \sin\left(\frac{x}{2}\right) \right) \\ &= \left(\lim_{t \rightarrow 0} \frac{\sin(t)}{t} \right) \left(\lim_{t \rightarrow 0} \sin(t) \right) \quad \text{let } t = \frac{x}{2} \\ &= -(1)(0) = 0. \end{aligned}$$

Limits Involving Infinity

Limits at Infinity

- ▶ ∞ is a symbol, it is not a number.
- ▶ The symbol $x \rightarrow \infty$ means when x becomes large.
- ▶ $\lim_{x \rightarrow \infty} f(x)$ describes the behavior of the function $f(x)$ when x grows continuously.
- ▶ Similarly $\lim_{x \rightarrow -\infty} f(x)$ describes the behavior of the function as we move to the left of the graph of $f(x)$.

Example

Let $f(x) = \frac{1}{x}$, $x \neq 0$.

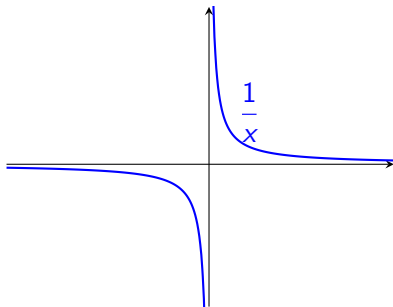
When $x \rightarrow +\infty$, $\frac{1}{x}$ becomes very small, but remains positive,
 $\frac{1}{x} \rightarrow 0^+$.

When $x \rightarrow -\infty$, $\frac{1}{x}$ again becomes small, but is negative, $\frac{1}{x} \rightarrow 0^-$.

So, we write

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0^+$$

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0^-$$



Definitions

1. $f(x)$ has limit L as x approaches infinity and write

$$\lim_{x \rightarrow \infty} f(x) = L$$

if, as x becomes positively large, $f(x)$ approaches L .

2. $f(x)$ has limit L as x approaches negative infinity and write

$$\lim_{x \rightarrow -\infty} f(x) = L$$

if, as x becomes negatively large, $f(x)$ approaches L .

Limit Laws for Limits $x \rightarrow \pm\infty$

Suppose

$$\lim_{x \rightarrow \pm\infty} f(x) = L, \quad \text{and} \quad \lim_{x \rightarrow \pm\infty} g(x) = M.$$

(i) (Sum/Difference rule): $\lim_{x \rightarrow \pm\infty} (f(x) \pm g(x)) = L \pm M$

(ii) (Scalar multiple rule): $\lim_{x \rightarrow \pm\infty} (k \cdot f(x)) = k \cdot L$

(iii) (Product rule): $\lim_{x \rightarrow \pm\infty} (f(x) \cdot g(x)) = L \cdot M$

(iv) (Quotient rule): $\lim_{x \rightarrow \pm\infty} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$

(v) (Power rule): $\lim_{x \rightarrow \pm\infty} [f(x)]^n = L^n, \quad n \in \mathbb{Z}^+$

(vi) (Root rule): $\lim_{x \rightarrow \pm\infty} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{\frac{1}{n}}, \quad n \in \mathbb{Z}^+, \text{ and if } n \text{ is even, we assume } L \geq 0.$

Example

1. Find $\lim_{x \rightarrow \infty} \frac{5x+1}{x}$.

$$\frac{5x+1}{x} = \frac{5x}{x} + \frac{1}{x} = 5 + \frac{1}{x}. \text{ So,}$$

$$\lim_{x \rightarrow \infty} \frac{5x+1}{x} = \lim_{x \rightarrow \infty} \left(5 + \frac{1}{x} \right) = 5 + \lim_{x \rightarrow \infty} \frac{1}{x} = 5.$$

2. Find $\lim_{x \rightarrow \infty} \frac{1}{x^2}$.

$$\lim_{x \rightarrow \infty} \frac{1}{x^2} = \left(\lim_{x \rightarrow \infty} \frac{1}{x} \right)^2 = 0^2 = 0.$$

Limits at Infinity of Rational Functions

Consider a rational function

$$\frac{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0}.$$

► Case 1. If $n = m$, then dividing throughout by x^n ,

$$\lim_{x \rightarrow \infty} \frac{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0}{b_n x^n + b_{n-1} x^{n-1} + \cdots + b_1 x + b_0} = \lim_{x \rightarrow \infty} \frac{a_n + \frac{a_{n-1}}{x} + \cdots + \frac{a_1}{x^{n-1}} + \frac{a_0}{x^n}}{b_n + \frac{b_{n-1}}{x} + \cdots + \frac{b_1}{x^{n-1}} + \frac{b_0}{x^n}} = \frac{a_n}{b_n}.$$

Limits at Infinity of Rational Functions

Consider a rational function

$$\frac{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0}.$$

► Case 2. If $n > m$, then dividing throughout by x^m ,

$$\lim_{x \rightarrow \infty} \frac{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0} = \lim_{x \rightarrow \infty} \frac{a_n x^{n-m} + a_{n-1} x^{n-m-1} + \cdots + \frac{a_1}{x^{m-1}} + \frac{a_0}{x^m}}{b_m + \frac{b_{m-1}}{x} + \cdots + \frac{b_1}{x^{m-1}} + \frac{b_0}{x^m}} = \pm \infty$$

since the denominator approach b_m and the numerator approach $\pm \infty$, depending on whether $\frac{a_n}{b_m} > 0$
or $\frac{a_n}{b_m} < 0$.

Limits at Infinity of Rational Functions

Consider a rational function

$$\frac{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0}.$$

► Case 3. If $n < m$, then throughout by x^m ,

$$\lim_{x \rightarrow \infty} \frac{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0} = \lim_{x \rightarrow \infty} \frac{\frac{a_n}{x^{m-n}} + \frac{a_{n-1}}{x^{m-n+1}} + \cdots + \frac{a_1}{x^{m-1}} + \frac{a_0}{x^m}}{b_m + \frac{b_{m-1}}{x} + \cdots + \frac{b_1}{x^{m-1}} + \frac{b_0}{x^m}} = 0$$

Examples

1. $\lim_{x \rightarrow \infty} \frac{5x^2 + 8x - 3}{3x^2 + 2}$

$$\lim_{x \rightarrow \infty} \frac{5x^2 + 8x - 3}{3x^2 + 2} = \lim_{x \rightarrow \infty} \frac{5 + \frac{8}{x} - \frac{3}{x^2}}{3 + \frac{2}{x^2}} = \frac{5}{3}.$$

2. $\lim_{x \rightarrow \infty} \frac{11x + 2}{2x^3 - 1}$

$$\lim_{x \rightarrow \infty} \frac{11x + 2}{2x^3 - 1} = \lim_{x \rightarrow \infty} \frac{\frac{11}{x^2} + \frac{2}{x^3}}{2 - \frac{1}{x^3}} = 0.$$

Horizontal Asymptotes

Definition

A line $y = b$ is a **horizontal asymptote** of the graph of a function $y = f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = b \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = b.$$

Example

1. $y = \frac{5}{3}$ is a horizontal asymptote of $\frac{5x^2 + 8x - 3}{3x^2 + 2}$.

2. $y = 0$ is a horizontal asymptote of $\frac{11x + 2}{2x^3 - 1}$.

Examples

1. Find $\lim_{x \rightarrow 0} e^{-\frac{1}{x^2}}$.

Let $t = \frac{1}{x^2}$. Then as $x \rightarrow 0$, $t \rightarrow \infty$. Hence,

$$\lim_{x \rightarrow 0} e^{-\frac{1}{x^2}} = \lim_{t \rightarrow \infty} e^{-t} = 0.$$

2. Find $\lim_{x \rightarrow \infty} (x - \sqrt{x^2 + 16})$.

$$x - \sqrt{x^2 + 16} = \frac{(x - \sqrt{x^2 + 16})(x + \sqrt{x^2 + 16})}{x + \sqrt{x^2 + 16}} = \frac{x^2 - (x^2 + 16)}{x + \sqrt{x^2 + 16}} = -\frac{16}{x + \sqrt{x^2 + 16}}. \text{ Hence,}$$

$$\lim_{x \rightarrow \infty} (x - \sqrt{x^2 + 16}) = \lim_{x \rightarrow \infty} \left(-\frac{16}{x + \sqrt{x^2 + 16}} \right) = -\lim_{x \rightarrow \infty} \frac{\frac{16}{x}}{1 + \sqrt{1 + \frac{16}{x^2}}} = 0.$$

Oblique Asymptotes

Let

$$f(x) = \frac{p(x)}{q(x)}$$

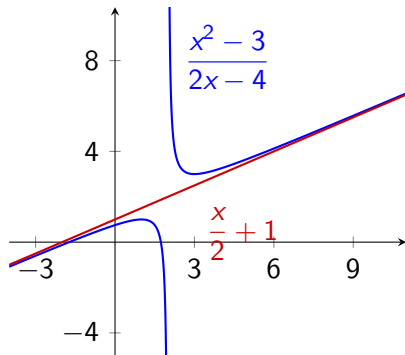
be a rational function such that $\deg(p) = \deg(q) + 1$. Then the graph has an oblique asymptote.

Example

Find the oblique asymptote of $f(x) = \frac{x^2 - 3}{2x - 4}$.

By long division,

$$\frac{x^2 - 3}{2x - 4} = \left(\frac{x}{2} + 1\right) + \left(\frac{1}{2x - 4}\right)$$

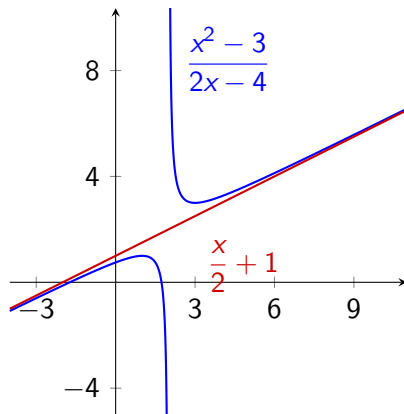


Oblique Asymptotes

Hence, as $x \rightarrow \pm\infty$, $\left(\frac{1}{2x-4}\right) \rightarrow 0$, and so

$$\frac{x^2-3}{2x-4} \rightarrow \left(\frac{x}{2} + 1\right).$$

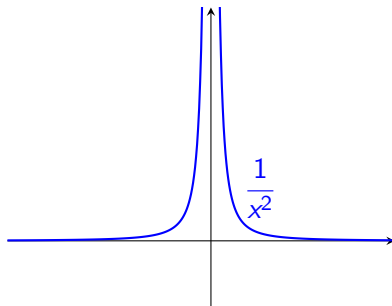
The line $y = \frac{x}{2} + 1$ is an (oblique) asymptote both to the left and to the right of $f(x)$.



Infinite Limits

Observe the graph of $\frac{1}{x^2}$, $x \neq 0$. When $x \rightarrow 0$, $\frac{1}{x^2}$ grows arbitrarily large without bound. So we write

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty.$$



Remark

- (i) Note that $\lim_{x \rightarrow c} f(x) = \infty$ **does not** mean that the limit exists. The limit **does not** exist.
- (ii) It describes the behavior of $f(x)$ as x approaches c , that is, $f(x)$ becomes **arbitrarily large and positive**.
- (iii) Similarly, $\lim_{x \rightarrow c} f(x) = -\infty$ means that $f(x)$ becomes **arbitrarily large and negative** as x approaches c .

Definition

1. $f(x)$ approaches infinity as x approaches c , and write

$$\lim_{x \rightarrow c} f(x) = \infty,$$

if for the graph of $f(x)$ grows arbitrarily large without bound as x approaches c .

2. $f(x)$ approaches negative infinity as x approaches c , and write

$$\lim_{x \rightarrow c} f(x) = -\infty,$$

if for the graph of $f(x)$ grows arbitrarily large and negative without bound as x approaches c .

Example

1. Find $\lim_{x \rightarrow 1} \frac{1}{(x-1)^2}$.

Let $t = x - 1$. Then $t \rightarrow 0$ as $x \rightarrow 1$. Hence,

$$\lim_{x \rightarrow 1} \frac{1}{(x-1)^2} = \lim_{t \rightarrow 0} \frac{1}{t^2} = \infty$$

2. Find $\lim_{x \rightarrow 2} \frac{2-x}{(x-2)^3}$.

$\frac{2-x}{(x-2)^3} = -\frac{x-2}{(x-2)^3} = -\frac{1}{(x-2)^2}$. Let $t = x - 2$, then $t \rightarrow 0$ as $x \rightarrow 2$. Hence,

$$\lim_{x \rightarrow 2} \frac{2-x}{(x-2)^3} = -\lim_{x \rightarrow 2} \frac{1}{(x-2)^2} = -\lim_{t \rightarrow 0} \frac{1}{t^2} = -\infty.$$

Example

3. Find $\lim_{x \rightarrow -\infty} \frac{2x^5 - 6x^4 + 1}{3x^2 + x - 7}$.

$$\frac{2x^5 - 6x^4 + 1}{3x^2 + x - 7} = \frac{2x^3 - 6x^2 + \frac{1}{x^2}}{3 + \frac{1}{x} - \frac{7}{x^2}}. \text{ Hence,}$$

$$\lim_{x \rightarrow -\infty} \frac{2x^5 - 6x^4 + 1}{3x^2 + x - 7} = \lim_{x \rightarrow -\infty} \frac{2x^3 - 6x^2 + \frac{1}{x^2}}{3 + \frac{1}{x} - \frac{7}{x^2}} = -\infty.$$

Vertical Asymptotes

A line $x=a$ is a **vertical asymptote** of the graph of a function $y=f(x)$ if either

$$\lim_{x \rightarrow a^+} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \rightarrow a^-} f(x) = \pm\infty.$$

Example

1. $x = 0$ is a vertical asymptote of $\frac{1}{x}$.

2. $x = -2$ is a vertical asymptote of

$$\frac{x+3}{x+2}.$$

Continuous Functions

Introduction

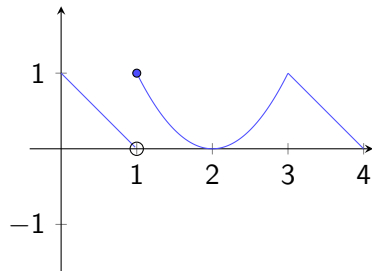
Recall that for any algebraic function $f(x)$ and $c \in D$ in the domain of f

$$\lim_{x \rightarrow c} f(x) = f(c).$$

This establishes that every algebraic function is continuous on its domain.

Assume that $\lim_{x \rightarrow c} f(x) = f(c)$.

- ▶ As x approaches c , the values of $f(x)$ approach $f(c)$.
- ▶ Hence, in a sufficiently small neighborhood of c , the graph of f has no jumps, gaps, or breaks. The graph passes smoothly through $x = c$, expressing the continuity of f at that point.



Continuous at a Point

Let $c \in \mathbb{R}$ be either in an interior point or an endpoint of an interval in the domain of $f(x)$. The function f is continuous at c if

$$\lim_{x \rightarrow c} f(x) = f(c).$$

Continuity Test

A function $f(x)$ is **continuous** at a point $x = c$ in its domain D if and only if it meets the following conditions.

1. $\lim_{x \rightarrow c} f(x)$ **exists**, that is, $f(x)$ has a limit at c .
2. $\lim_{x \rightarrow c} f(x) = f(c)$, that is, the limit is equal to the function value.

Examples

1. Let $f(x) = \lfloor x \rfloor$. For any integer n ,

$$\lim_{x \rightarrow n^-} f(x) = n - 1 \quad \text{and} \quad \lim_{x \rightarrow n^+} f(x) = n.$$

The limit does not exist, hence the function is not continuous. This is known as a jump discontinuity.

2. Let

$$f(x) = \begin{cases} x + 1 & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Then

$$\lim_{x \rightarrow 0} f(x) = 1 \neq 0 = f(0).$$

So, the function is not continuous at $x = 0$. This is known as a removable discontinuity.

Examples

3. Let

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Then $\lim_{x \rightarrow 0} f(x)$ is undefined since the function oscillates up and down between 1 and -1 infinitely many times as $x \rightarrow 0$. This is known as a oscillating discontinuity.

Continuous Functions

Definition

A function f is said to be **continuous** on an **interval** $I \subseteq D$ contained in its **domain** if it is continuous at **every** $x \in I$.

A function f is said to be **continuous** if it is continuous at **every** $x \in D$ in its **domain**.

Remark

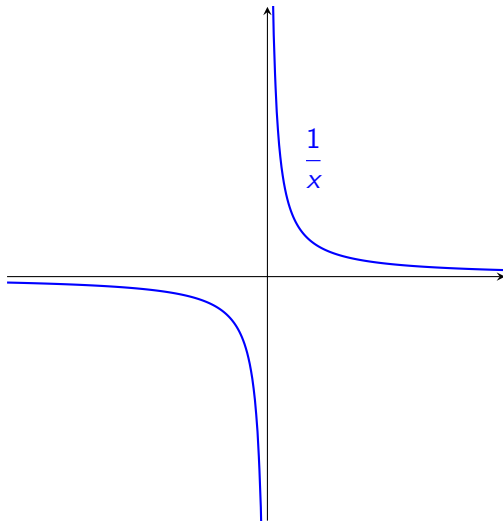
Remember the definition of a function comes with its domain. So, continuity is a property of the function. If we change the domain, we change the function, and this may change its continuity property.

Question

Is the function

$$f(x) = \frac{1}{x}$$

continuous?



Algebraic Operations Preserve Continuity

Theorem

Suppose f and g be *continuous* at $x = c$. Then the following *algebraic combinations* are *continuous* at $x = c$.

1. Sum/Difference. $f \pm g$
2. Scalar multiple. $k \cdot f$ for any $k \in \mathbb{R}$
3. Product. $f \cdot g$
4. Quotient. $\frac{f}{g}$ if $g(c) \neq 0$
5. Power. f^n for positive integer $n \in \mathbb{Z}^+$.
6. Roots. $\sqrt[n]{f}$ where n is a positive integer, $n \in \mathbb{Z}^+$. If n is even, we assume that $f(x) \geq 0$ for x in an interval containing c .

Algebraic Functions are Continuous

Corollary

All algebraic functions are continuous.

Trigonometric Functions are Continuous

Theorem

All trigonometric functions are continuous wherever they are defined.

Students may verify the continuity of the trigonometric functions by examining their graphs.

- ▶ <https://www.geogebra.org/calculator>
- ▶ $\sin(x)$
- ▶ $\cos(x)$
- ▶ $\tan(x)$
- ▶ $\sec(x)$
- ▶ $\operatorname{cosec}(x)$
- ▶ $\cot(x)$

Why Trigonometric Functions Are Continuous

The continuity of the trigonometric functions can be understood from their geometric definitions.

- ▶ On the unit circle, the point corresponding to angle x has coordinates $(\cos x, \sin x)$. Both coordinates vary smoothly as the point moves around the circle.
- ▶ The function $\sin x$ gives the y -coordinate of this point, and $\cos x$ gives the x -coordinate. Since the point moves without jumps, $\sin x$ and $\cos x$ change without jumps and are therefore continuous.
- ▶ The remaining trigonometric functions are formed by quotients:

$$\begin{aligned}\tan x &= \frac{\sin x}{\cos x} \\ \sec x &= \frac{1}{\cos x} \\ \csc x &= \frac{1}{\sin x} \\ \cot x &= \frac{\cos x}{\sin x}\end{aligned}$$

- ▶ They are continuous wherever their denominators are nonzero.

Limits of Continuous Functions

Theorem

If $\lim_{x \rightarrow c} f(x) = a$ and g is continuous at a , then $\lim_{x \rightarrow c} g \circ f(x) = g(a)$.

Equivalently,

$$\lim_{x \rightarrow c} g \circ f(x) = g\left(\lim_{x \rightarrow c} f(x)\right).$$

That is, for a continuous function g , we may bring the limit inside the function.

Furthermore, if f is also continuous at c , then

$$\lim_{x \rightarrow c} g \circ f(x) = g\left(\lim_{x \rightarrow c} f(x)\right) = g(f(c)).$$

Thus the composition of continuous functions is continuous at the point of composition.

Composition of Continuous Functions

Corollary

If f is *continuous* at c and g is *continuous* at $f(c)$, then the *composition* $g \circ f$ is *continuous* at c .

Corollary (Composition of continuous functions is continuous)

If f and g are *continuous* on their respective domains, then the composition $f \circ g$ is *continuous* on its natural domain.

Examples

1. The function $f(x) = \sqrt{x^2 - 2x - 5}$ is continuous on its natural domain, since algebraic functions and the square-root function are continuous on their respective domains.
2. The function $f(x) = \left| \frac{x-2}{x^2-2} \right|$ is continuous on its natural domain $x \neq \pm 2$. The modulus function $|\cdot|$ is continuous everywhere, and the rational expression $\frac{x-2}{x^2-2}$ is continuous wherever its denominator is nonzero.
3. Consider $f(x) = \left| \frac{x \sin x}{x^2 + 2} \right|$. The function $\sin x$ is continuous everywhere, and hence $x \sin x$ is continuous everywhere. The denominator $x^2 + 2$ is continuous and strictly positive everywhere, so $\frac{1}{x^2 + 2}$ is continuous everywhere. Therefore the function $f(x)$ is continuous for all real x .

Intermediate Value Theorem

Consider the polynomial

$$p(x) = x^3 - x^2 - 2x + 2.$$

Observe that

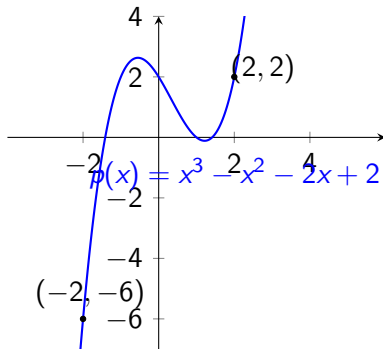
$$p(-2) = -6 < 0 \quad \text{and} \quad p(2) = 2 > 0.$$

By **continuity**, it must be that there is a $c \in (-2, 2)$ such that

$$p(c) = 0.$$

Theorem (Intermediate value theorem)

Suppose f is a **continuous function** on a **closed interval** $[a, b]$. Then for **any** y_0 **between** $f(a)$ and $f(b)$, **there exists** a $c \in (a, b)$ such that $f(c) = y_0$



Example

1. Show that the polynomial $x^3 - x - 1$ has a root between 1 and 2.

Let $p(x) = x^3 - x - 1$. Then $p(1) = 1^3 - 1 - 1 = -1 < 0$ and $p(2) = 2^3 - 2 - 1 = 5 > 0$. So, $p(x)$ must have a root between 1 and 2.

2. Prove that the two graphs $y = \sqrt{2x+5}$ and $y = 4 - x^2$ intersect.

Consider the function $f(x) = \sqrt{2x+5} - (4 - x^2) = \sqrt{2x+5} - 4 + x^2$. Then $f(c) = 0$ if and only if the two graphs intersect at $x = c$.

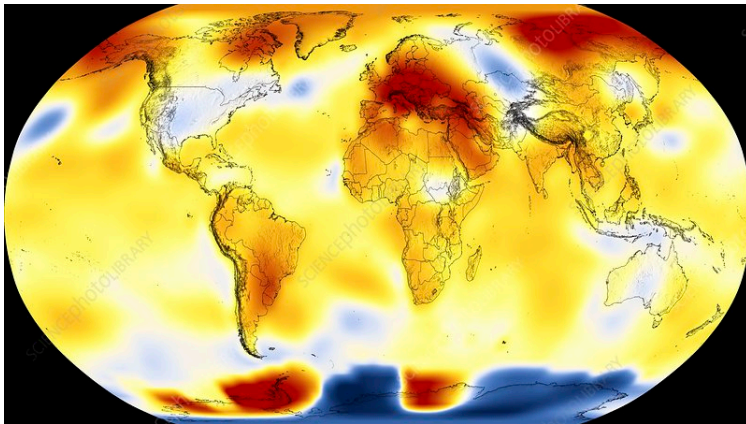
- ▶ $f(-2) = \sqrt{1} = 1 > 0$,
- ▶ $f(0) = \sqrt{5} - 4 < 0$, and
- ▶ $f(2) = \sqrt{9} = 3 > 0$

tells us that the graphs intersect in the intervals $(-2, 0)$ and $(0, 2)$.

Exercise. Show that all odd polynomials has at least one real root.

Application

There are always two points on opposite sides of the Earth with the exact same temperature.



<https://youtu.be/5Px6fajpSio>